

Power Requirements

First and foremost, your TPJ project must remain **electrically safe** throughout its full usage. There are two main types of power supplies for projects: AC or DC. For TPJ, **use a DC supply**. Within the DC supply options, you can learn about “linear” or “switching” power supplies.

Linear Power Supplies

For linear power supplies, a simple setup would be to use a 9 VDC wall adapter, install a barrel connector on your project enclosure, and then have a linear regulator providing the step-down voltage you need (eg. 5 VDC). Pros: It’s cheap and easy to install. Cons: It’s energy inefficient.

Linear Voltage Regulators (LM7805) | AO #17

<https://youtu.be/OAoEWaGtQjs>

By: AddOhms (@AddOhms)

LM7805 has 5 VDC output. LM7809 has 9 VDC output. LM7812? You guessed it: 12 VDC output.

Switching Power Supplies

Typical modules that fall into this category are the Buck Converter, Boost Converter, and Buck-Boost Converter. You may also know them as “Step-Down” and “Step-Up” voltage converters. Pros: Energy efficient. Cons: Expensive and often have a ripple (due to switching nature).

Switching Voltage Regulator (Buck, Boost) Introduction | AO #18

<https://youtu.be/CPvA-uS0dNY>

By: AddOhms (@AddOhms)

Buck converter vs. linear voltage regulator - practical comparison

<https://youtu.be/qiGRrODKJSE>

By: ElectronFun.com (@ElectronFunCom)

Is There a Better Option?

Suppose your solenoid output valve runs on 12 VDC. Your 5 VDC microcontroller cannot supply that voltage/current (and you’ll probably brick it if you try!). Try designing a system that has a 12 VDC input wall adapter, a buck converter to step down the voltage to 6 VDC on your PCB, then have one LM7805 linear regulator installed. In this way, you’ll have both 5 VDC and 12 VDC power lines available on your PCB, plus reduced heat dissipation. Segregate their own areas.

From the microcontroller, you can design one of its GPIO pins to turn on a relay to turn on the solenoid or, better yet, insert a transistor (eg. FET + diode) that connects to the load output.

Connecting Lights, Motors and Solenoids to Microcontrollers

<https://youtu.be/epEGN-7GsPs>

By: Espruino (@Espruino)

The combination of using a buck converter with a linear regulator uses the advantage of each: the former steps down the voltage safely, where the latter smooths any ripple on the output.